Causal Machine Learning

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1 Conditional Exchangeability

We have X for eligibility status ($X = 1$ if eligible and $X = 0$ otherwise), Y for accumulated financial assets (outcome), and V income. We reason that employers will more likely offer a 401(k) to their employees who earn more as compared to those wo earn less. Therefore, V affects X. Furthermore, a high income allows one to save more over time resulting in more accumulated financial assets, so V also affects Y.

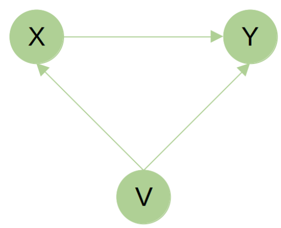


Figure 1: Directed acyclic graph (DAG) showing relationships between X, Y, and V.

Eligibility status for 401(k) is not assigned randomly to each employee, but is influenced by several factors, among which income V. Therefore, eligibile and non-eligible people are not comparable. If we believe that income is the only factor influencing eligibility, then conditioning on income makes eligibile and non-eligible people more comparable. We can state $Y^x \perp\!\!\!\perp X | V$.

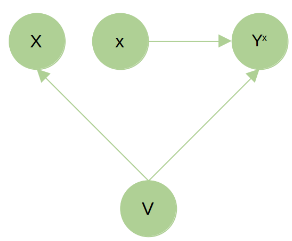


Figure 2: DAG with counterfactuals.

2 AIPW and TMLE

For both AIPW and TMLE, non-parametric models using SuperLearner were fitted to estimate the nuisance parameters. Five-fold cross-fitting was applied. AIPW gives an estimate for the average treatment effect in the treated (ATT) of 8,680.89 (95% CI: [3,838.80, 13522.98]). TMLE gives an estimate for the ATT of 5960.71 (95% CI: [1,072.32, 10,849.09]). To estimate the ATT when no covariate adjustment is made, we calculate an estimate for ATT as follows:

$$
\begin{aligned}
E(Y^1 - Y^0|A=1) &= E(Y^1|A=1) - E(Y^0|A=1) \\
&= E(Y|A=1) - E(Y^0|A=0) \\
&= E(Y|A=1) - E(Y|A=0)
\end{aligned}
$$

Because we have $Y^a \perp\!\!\!\perp A$, we can state that $E(Y^a|A=0) = E(Y^a|A=1)$. This gives us an estimate for the ATT, without covariate adjustment, of 19,559.34. We see that the estimate for the ATT both with covariate adjustment (AIPW and TMLE) and without covariate adjustment gives a positive effect, which is in line with the expectation. However, when adjusting for covariates, the effect is less pronounced than when not adjusting for covariates, which indicates that when comparing similar people, the effect of eligibility for 401(k) is less pronounced. Furthermore, we also see that, although there is a positive effect with covariate adjustment, the effect is not that large compared to the (absolute) values of the accumulated financial assets. The ATT can be interpreted as follows: when we look at people that are eligible for 401(k) (i.e, the treated/exposed) and assume they are no longer eligible for 401(k), then, on average, their accumulated financial assets would be 8,680.89 USD less (AIPW estimate) if their eligibility for 401(k) would be taken away. We opted to only look at those who were eligible, i.e., average treatment effect in the treated, because then we would see the impact of eligibility for 401(k) for those already eligible and really assess the impact this eligibility has on the accumulated assets.

When checking the estimates of the propensity scores, we do not see any violation of the positivity assumption nor excessive weights:

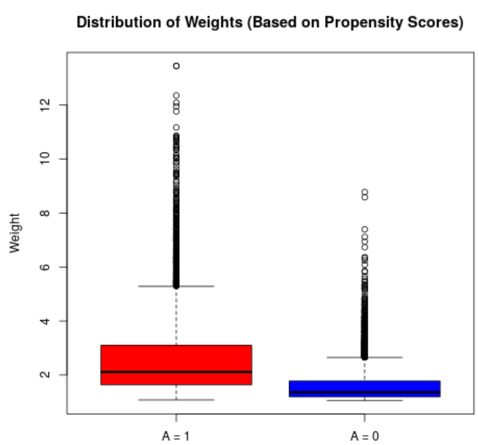


Figure 3: Box plots showing distribution of weights based on propensity scores.

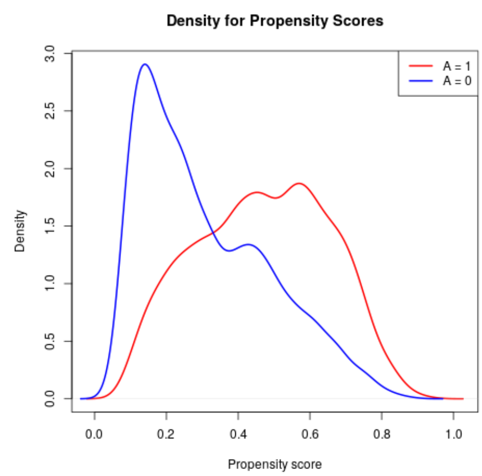


Figure 4: Distribution of propensity scores.

3 Estimate θ

We have the following estimator:

$$
0 = E[(A - E(A|L))(Y - E(Y|L)) - \hat{\theta}] \leftrightarrow \hat{\theta} = E[(A - E(A|L))(Y - E(Y|L))]
$$

This estimator yields an estimate of -44.88131. We apply the efficient influence curve to each observation. We then use these EIC values to yield a standard error as follows:

$$
\frac{1}{\sqrt{n}} \cdot \sqrt{Var(EIC)}
$$

We obtain a standard error of 269.75. If we have the following null and alternative hypothesis:

$$H_0 : \theta = 0 \text{ vs } H_a : \theta \neq 0$$

then we obtain a p-value of $p = 0.8679$ for a 2-sided test. We can therefore not reject H_0. The covariance between the outcome Y and the exposure A conditioning on the confounders L is 0, which seems to indicate that there is no covariance (nor correlation).

4 R-learner

4.1 Approach

An rlearner approach was used to estimate the conditional average treatment effect of the eligibility for enrolling in a 401(k) plan on the net financial assets in function of age, income and education, adjusting for other confounders. The model used for this task was a random forest model that uses honest trees, implemented in the *causal_forest()* function of the *grf* package. To reduce overfitting, 5-fold cross-fitting was implemented and some parameters of the random forest model where tuned.

4.2 Results

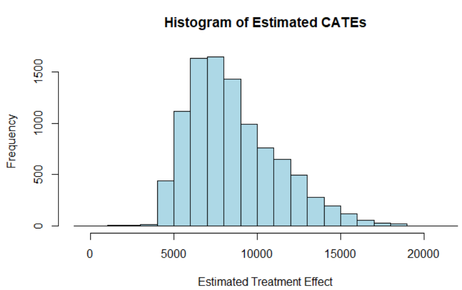


Figure 5: Distribution of CATE estimates.

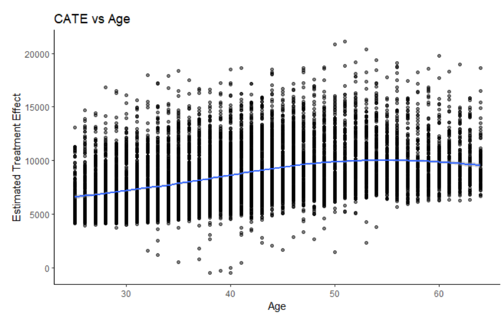


Figure 6: CATE vs. Age.

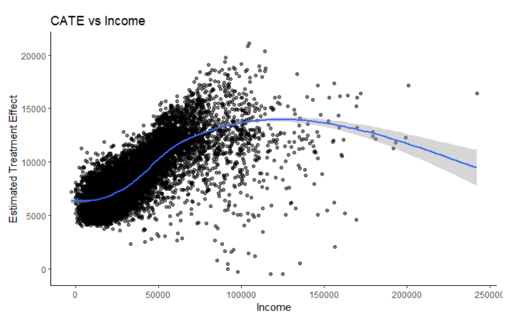


Figure 7: CATE vs. Income.

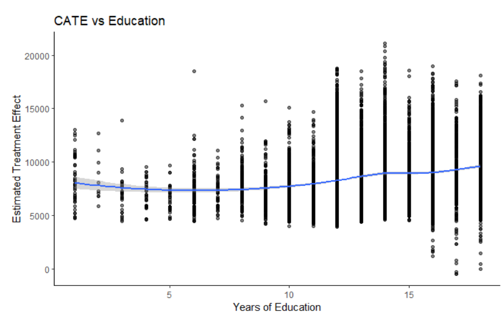


Figure 8: CATE vs. Education.

4.3 Interpretation

Treatment effects for all individuals range from approximately -450 to 21,000 with most of the individuals in the study showing a treatment effect of 8046 (median) (Figure 5).

The marginal effect of age shows a slight increase in treatment effect for increasing age. This indicates that the difference between net financial assets of a person that is eligible for the 401(k) plan compared to someone who isn't, is larger for older people than for younger people (Figure 6).

Income appears to have a strong positive correlation with the treatment effect. For people with an income of 100,000 US dollars, the effect of eligibility is approximately 150% that of people with an income of 50,000 US dollars. For people with very high incomes ($> 100{,}000\$$) no real conclusions can be drawn as here the number of individuals is limited and one must be weary of extrapolation errors (Figure 7).

In general the years of education of people does not seems to affect the estimates to a large extent (Figure 8).

5 Counterfactual Prediction of Net Assets under Eligibility

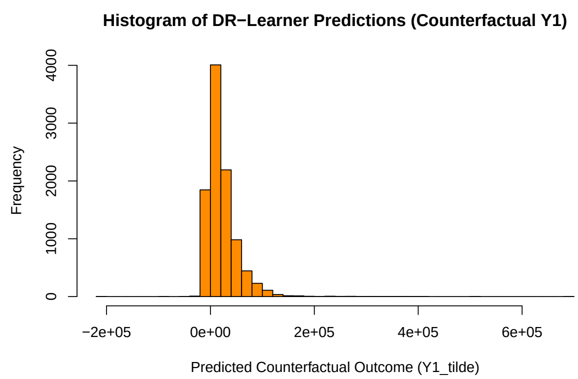
5.1 Approach

DR-Learner with five-fold cross fitting and SuperLearner (GLM, random forest and xgboost)

Prediction Error and 95% CI:

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} \frac{A_i}{\hat{p}_i}(Y_i - \tilde{Y}_i^1)^2, \quad \text{CI} = \hat{\theta} \pm 1.96 \text{ SE} \quad \text{with} \quad \text{SE} = \frac{\text{SD}(\varphi)}{\sqrt{n}}$$

5.2 Results



Prediction Error	95% CI
13,129,676,686.54	[7,915,274,814.41, 18,344,078,558.67]

5.3 Interpretation

Predictions range within $(-165,884.6, 618,810.3)$. Positivity is not violated, with propensity scores ranging within $(0.05, 0.95)$. The maximum IPW weight of 14.6 and standard deviation of 1.2 signal that individuals get upweighted but not radically so. The top five predictions with largest prediction error explain 37.89 percent of prediction error. They occur in case of large negative and positive net assets.

6 Division of Tasks

The questions were divided as follows:

- Q1 - Q3: Frederick De Baene
- Q4: Jonas Geysen
- Q5: Antoon Goderis